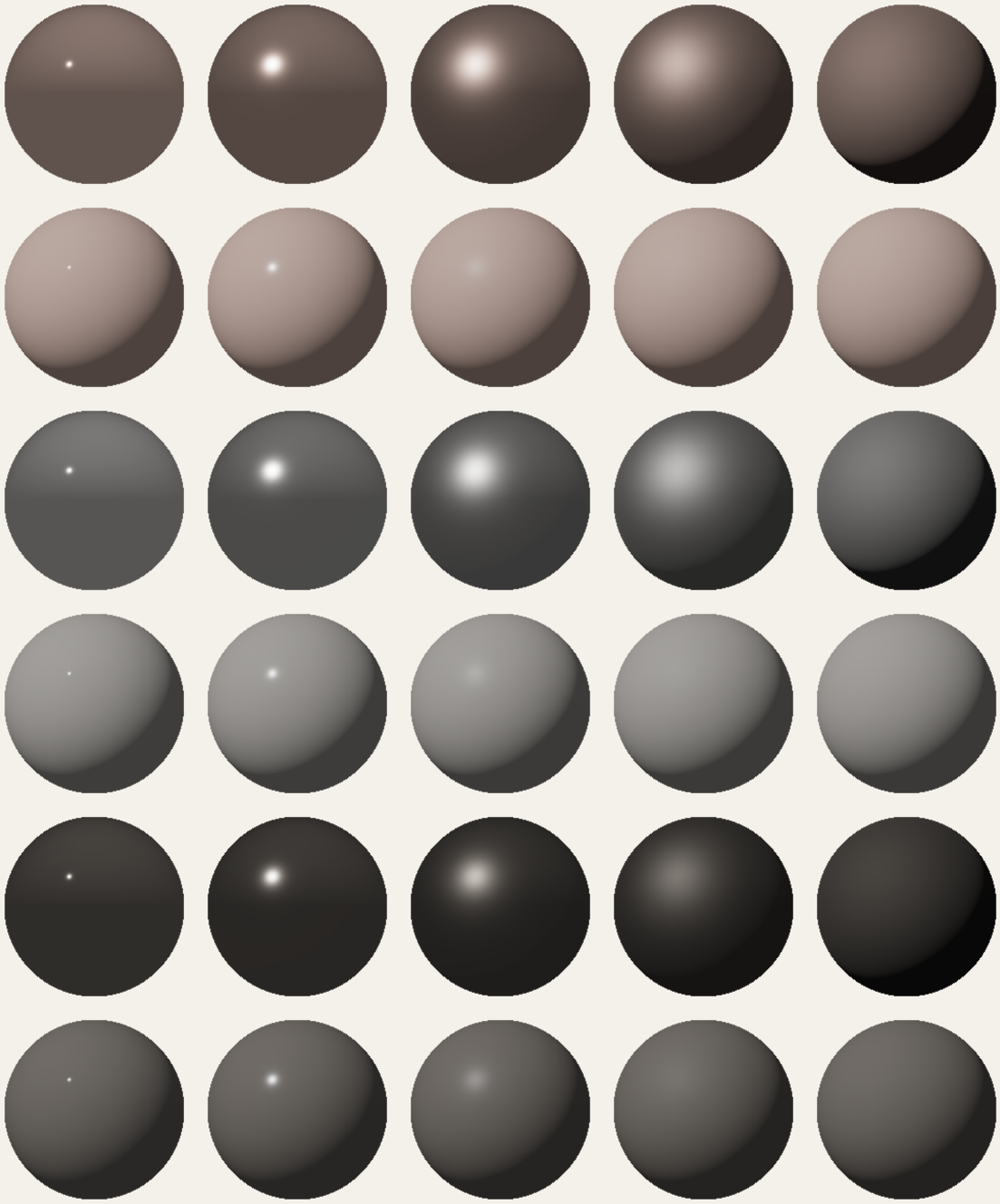


3DS MAX · V-RAY · CORONA · VERSION 5 · NOVEMBER 2026

Measured Light

The complete plan for a hyper-realistic material-creation plugin for 3ds Max

Perfect colour, roughness, metalness, normal, height, opacity, glass and translucency maps — built once from measured data and emitted as native **VRayMtl** and **CoronaPhysicalMtl**. The plugin makes materials; you render them.

PART I	Product plan — what it is and why it wins	03
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12	map types, each with a measured source
150	measured presets at v1.0
10 wk	to first revenue
30 / 46 wk	team of three / solo

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03 Positioning — Chaos's home turf
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Confidence tags used throughout the full spec: [Certain] verified against primary sources or standard physics · [Likely] strong inference, confirm in Phase 0 · [Guessing] estimate, to be calibrated. Durations, prices and targets are estimates.

You are on Chaos's home turf

Chaos makes V-Ray and Corona, and ships 3,100+ scanned materials inside both through Cosmos (Corona 12.2, Apr 2025; V-Ray 7.1, May 2025). A plain material library loses.

01

Own samples, same day

Capture the client's exact floor, stone or fabric in-house. No public evidence of a Chaos service that scans customer samples.

02

One source, two renderers

Studios mix V-Ray and Corona. One canonical material emits both natively, matched within ΔE 2.0 — and flagged if not.

03

Fix what's already there

Archviz scenes are full of broken downloaded materials. A scene-wide inspector with one-click fixes sells from week 10.

Confidence: Chaos Scans facts [Certain, verified Nov 2026]; the size of each opening [Likely]. Opening 03 is the fastest revenue because it needs no camera rig and no AI.

Three modules, shipped in order

Each module funds and de-risks the next. The validator built for the Inspector is what makes the Builder and the AI trustworthy.

A Material Inspector

MVP · PHASE 1

- Scans every VRayMtl, CoronaPhysicalMtl, legacy material and bitmap
- Flags data maps loaded as sRGB, fake metals, wrong IOR, glass without absorption, normal-map flips
- One-click undoable fixes + a delivery report

B Builder + Capture

PHASES 2–5

- Measured presets: metals (complex IOR), glass families, stone, skin
- Tripod RAW capture of a real sample → full map set
- Imports Substance / Megascans / Poliigon sets and fixes them
- Emits native VRayMtl and CoronaPhysicalMtl

C AI Generator

PHASE 6

- Text or photo → base, normal, height, roughness, metal
- Seamless, 2–4K, licence-clean training data
- Must pass the Inspector's rules before projection

Non-negotiable: every generated material uses only stock V-Ray, Corona and 3ds Max nodes, so scenes render on a farm or a client's machine without the plugin installed.

Inside 3ds Max, engine beside it

No GPL constraint in Max — the plugin can be closed and sold. The map engine runs separately: a 2 GB PyTorch install and Max's single UI thread don't mix. We make materials; the user renders.



Matrix to start: 3ds Max 2025–2027 · V-Ray 7 · Corona 13–14 · Windows x64. Max 2026 bundles Python 3.11.9 and OpenColorIO 2.4.1, so OCIO colour management is the baseline.

Distribution: Autodesk App Store or own store, as an application package (PackageContents.xml under ApplicationPlugins). Own licence server allowed.

One material, two renderers

The canonical material is translated into each renderer's native parameters. The translation is fitted by rendering, never assumed.

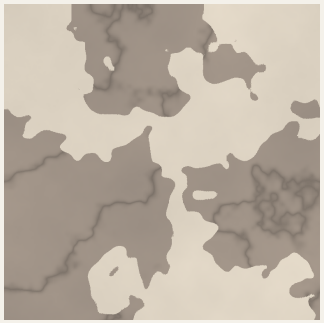
CHANNEL	V-RAY · VRayMtl	CORONA · CoronaPhysicalMtl	RULE
Base colour	Diffuse	Base color	sRGB, luminance 30/50–240
Roughness	GGX + Use roughness (reflection only)	Base roughness	set refraction glossiness = 1 - r
Metal	Metalness 1, base = metal colour	Complex IOR (n, k) or Edge Color	n,k from measured tables
Glass	Refract + IOR + fog colour	Refraction + IOR + volumetric absorption	measured absorption, always
SSS	Translucency: SSS / Volume	SSS mode or Volumetric mode	oa / os' fitted per renderer
Thin parts	Translucency / VRay2SidedMtl	Translucency / Thin shell	thickness baked per object
Normal	VRayNormalMap, flip set	CoronaNormal, flip G set	loaded linear, convention-tested
Tiling	VRayUVWRandomizer (stochastic)	CoronaUvwRandomizer	never on bricks or planks

Internal only. We fit this table once per V-Ray and Corona release so the same map set looks the same in both renderers. The user never runs a test — they just get matching materials.

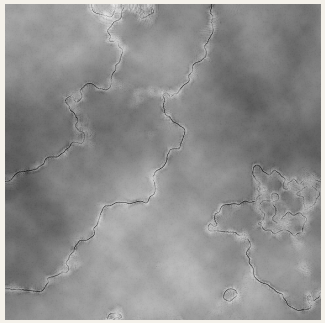
SAME MAPS · SAME LOOK

One structure signal, six maps

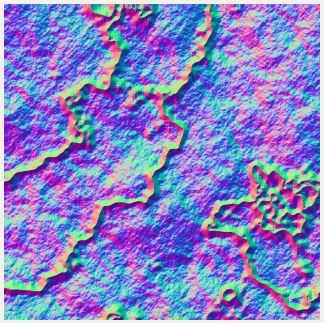
A scratch must change colour, roughness and height together. Independent maps are the first thing that makes a material read as CG.



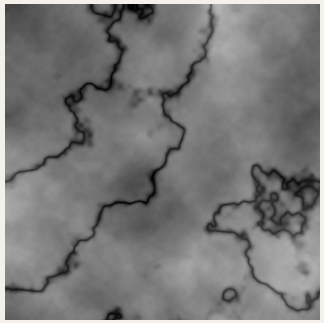
BASE COLOUR
sRGB · luminance 30/50–240



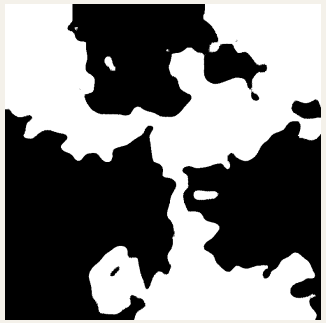
ROUGHNESS
linear 16-bit · 0–1 legal



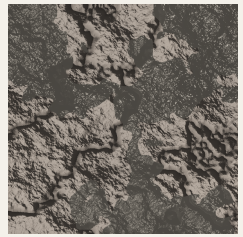
NORMAL
+Y MikkTSpace · = ∅height



HEIGHT
EXR · mm as unit attribute



METALLIC
bimodal ≥ 95 % at 0 or 1



SHADED RESULT · GGX, one light

01 Dielectric albedo

One scalar per pixel, never per channel: 30 sRGB tolerant / 50 strict, ceiling 240. Charcoal ≈ 39, fresh snow up to 249 on a whitelist.

02 Metal colour

Base colour = linear F0 from measured n,k under D65. v2's "≥ 180 sRGB per channel" rejected real gold.

03 Roughness

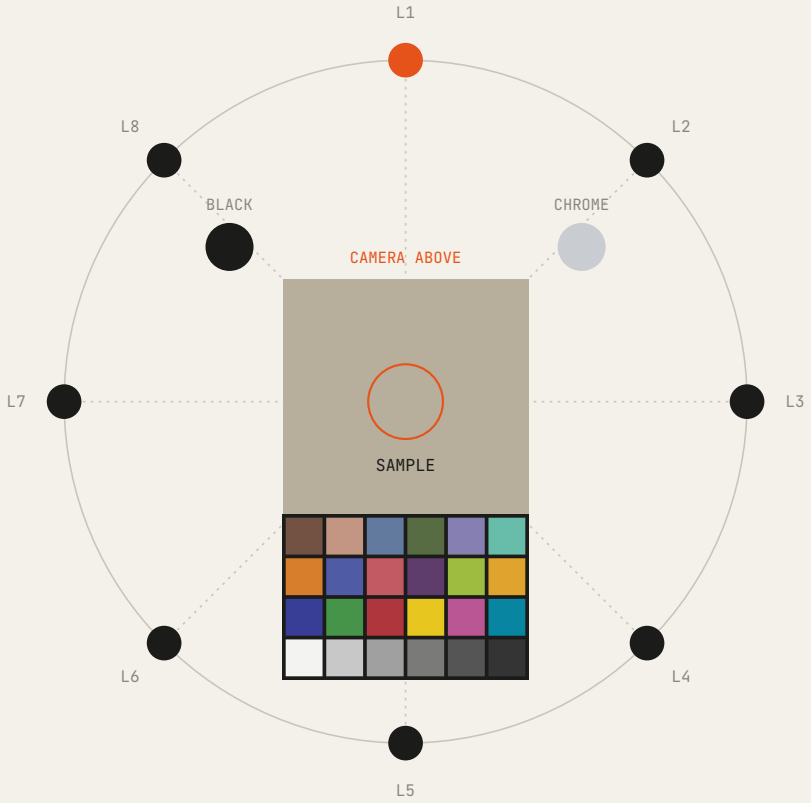
0 and 1 are legal (glass, chalk). Illegal: a spatially constant map. Variation tracks curvature, not height.

04 Subsurface

Store measured σa / σs′, never radii. Skin1 mean free path (1.30, 0.95, 0.67) mm — v2's values were a relabelled ratio.

A tripod, not a hand

Photometric stereo needs a fixed camera and a moving, polarised light. A phone's flash is bolted to its lens, so it cannot be the moving light.



TOP VIEW · 8 azimuths × 45° · elevation 45–60° · distance ≥ 3× sample

01 **Bayer RAW only**

JPEG, HEIC and Apple ProRAW are fused and tone-mapped; frame-to-frame they are not linear. Exposure, ISO, focus and white balance locked.

02 **Crossed + parallel pair per light**

Re-null the light's polariser on the black sphere at every position. One extra frame from the phone's own flash samples the specular peak.

03 **Polarisation split**

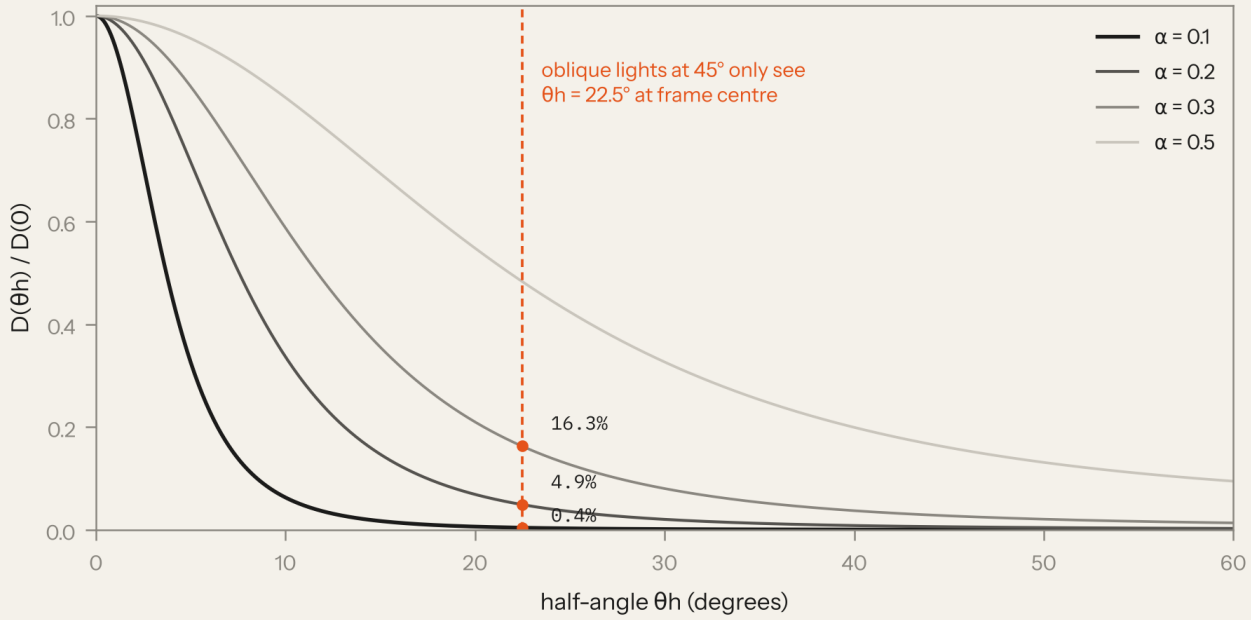
$I_{cross} = m \cdot D/2$ and $I_{par} = m \cdot (D/2 + S)$. So albedo $D = 2 \cdot I_{cross}/m$ and specular $S = (I_{par} - I_{cross})/m$, with m fixed from the checker greys. v2's subtraction left half the diffuse in the specular map.

04 **Near-light photometric stereo**

Each light is a calibrated point source with inverse-square fall-off; across a 30 cm sample irradiance varies ~3×. Integrate height with a Poisson solve, not periodic Frankot–Chellappa.

Why a phone cannot measure roughness per pixel

Lights at 45° elevation meet the centre of the sample at a 22.5° half-angle, where a glossy lobe has almost no energy.



The lobe is invisible to the oblique lights. At θh = 22.5° the GGX distribution returns 0.4 %, 4.9 % and 16 % of its peak for α = 0.1, 0.2 and 0.3. Anything glossier than α ≈ 0.3 sends essentially no specular signal to the 6–8 oblique frames, so a per-pixel fit is unconstrained.

What v3 does instead. Fit one α per material from the co-located flash frame — there the half-angle equals the view angle, so the highlight's fall-off traces the lobe from 0° to ~35°. Spatial variation comes from a stationarity prior and is tagged *prior-derived*, reported separately in the score.

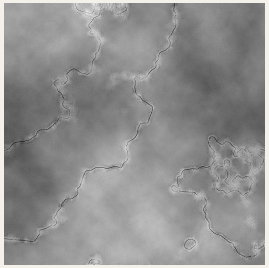
Every published phone pipeline does the same: Aittala 2015, Deschaintre 2018/2019, Adobe Sampler. True per-pixel lobe width needs a light dome with ≤ 7° spacing.

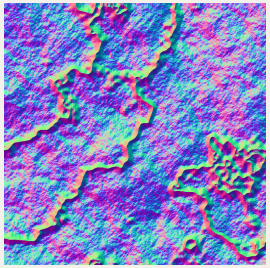
Perfect maps, checked as maps

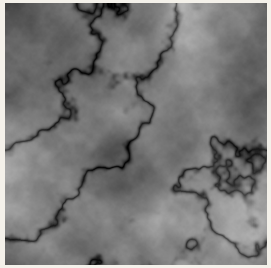
Every map is checked directly — colour, roughness, normal, height, metal, tiling, colour space — before it becomes a material. No rendering needed for the user.

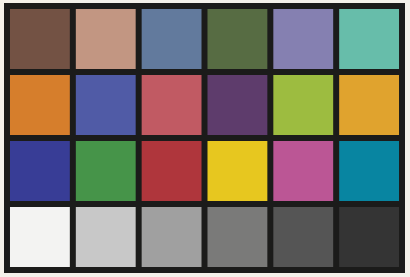
Bayer RAW › checker matrix › **Linear Rec.709 D65** › albedo ΔE ≤ 2 › roughness + metal › normal = ∂height › seamless tiling › colour space per map











ColorChecker in every capture: true colour, not camera colour

Colour, in two steps

Camera chain: mean ΔE2000 ≤ 2.0 over 24 patches. Material: ≤ 2.0 against a spectrophotometer at ≥ 3 spots. A checker-only test measures the camera, not the wood.

Every map, every time

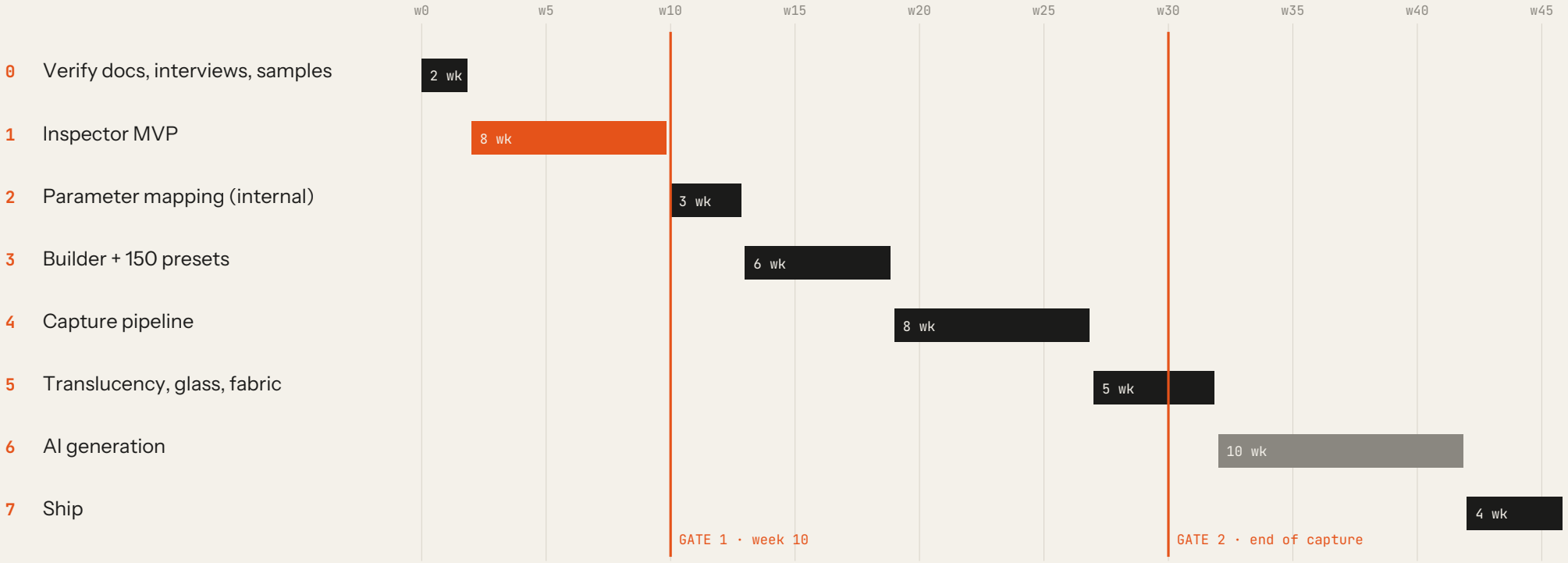
Ranges, roughness spread, normal-vs-height consistency, wrap seams and colour space are checked on each map set. Failures are fixed or flagged before the material is built.

Final proof

For flagship presets only, a blind “which is the photo?” test with archviz artists — run by us, once, as marketing proof.

Solo timeline: revenue at week 10

One developer, 46 weeks. The team-of-three version (30 weeks, Gate 2 at week 19) is in Part II. Durations are estimates.



Gate 1 — will studios pay? Inspector beta to 10 studios. Fewer than 5 willing to pay → rethink before building capture.

Gate 2 — does capture win? If it doesn't beat existing workflows on accuracy or turnaround for a customer's own sample, ship Inspector + Builder + presets and drop capture.

12-week version. Phase 0 + Inspector + a small Builder with ~40 measured presets for both renderers. A sellable product; capture and AI follow paying users.

What never ships

1. Native V-Ray and Corona nodes only — materials render without the plugin.
2. Roughness, gloss, metal, normal and height maps are always linear.
3. Never bake lighting or AO into base colour.
4. Never eyeball IOR, metal colour or SSS — every value has a source or a badge.
5. Never treat phone or AI roughness as measured.
6. Glass always carries a measured absorption colour.
7. Every material ships its validation report and renderer versions.
8. Every map passes its quality check before it becomes a material.

Biggest risks. Chaos ships the same features · renderer parameters change between versions (calibration suite in CI) · materials must work wherever the user renders (native nodes only) · GPU contention with the user's renders (engine paused while they render).

Still to confirm in Phase 0 week 1: official V-Ray 7 and CoronaPhysicalMtl parameter pages, Corona batch rendering, current Corona version, Chaos developer licences for CI.

The plan on one page

Three modules, eight phases, six milestones. The Inspector ships first and earns money while the harder modules are built.

WHAT

Material creation for 3ds Max: perfect colour, roughness, metal, normal, height, opacity, glass and translucency maps, built as native VRayMtl and CoronaPhysicalMtl.

WHO

Three engineers — plugin, imaging, maps/ML — plus a part-time material artist and the founder on product and sales. Solo is possible at 46 weeks.

WHEN

30 calendar weeks with three people. Revenue from week 10. Go / no-go at week 10 (will studios pay?) and week 19 (does capture beat existing workflows?).

HOW

Two-week sprints, a self-hosted Windows test machine with Max + V-Ray + Corona, every material through the same capture → maps → QA → build line.

MILESTONES · TEAM OF THREE · CALENDAR WEEKS



Five roles, clear ownership

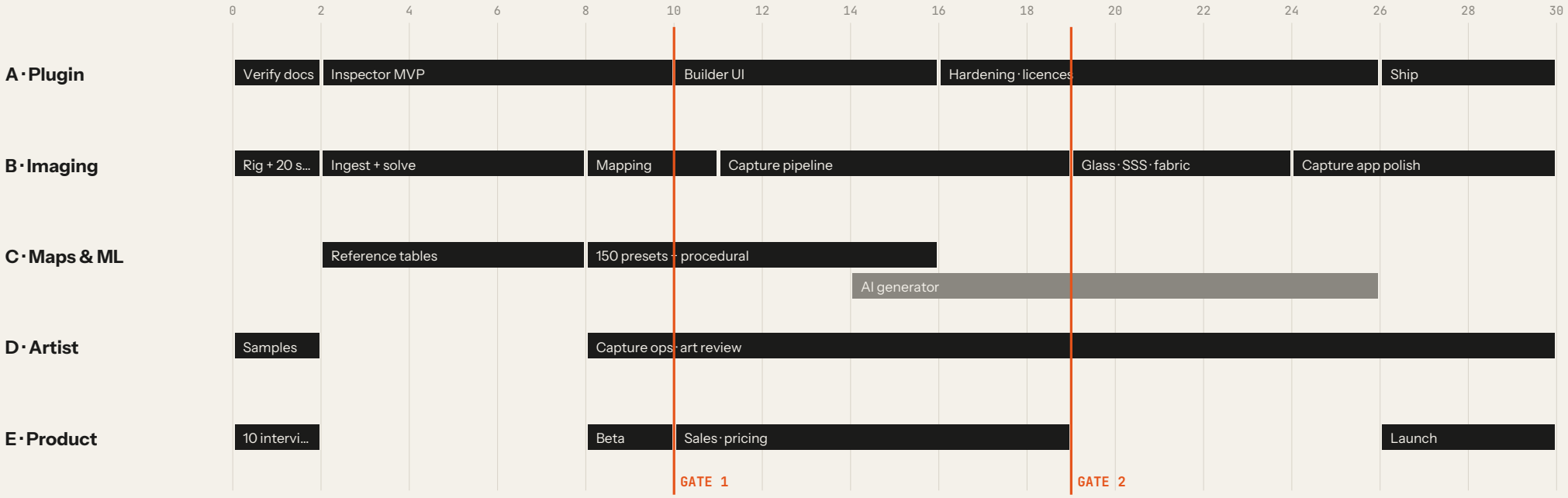
Hire for the three hard skills; borrow the rest. Each module has exactly one owner.

	ROLE	OWNS	MODULES
A	**Plugin engineer** 3ds Max · pymxs · PySide6	Inspector, Builder UI, material node construction, packaging, licensing, installer	INSPECTOR (A) · BUILDER UI · SHIP
B	**Imaging engineer** colour science · photometry	Capture rig, RAW ingest, polarisation split, photometric stereo, parameter mapping, glass & SSS	CAPTURE · MAPPING · TRANSLUCENCY
C	**Maps & ML engineer** PyTorch · procedural	Reference tables, procedural wear, tiling, preset library pipeline, AI generator	PRESETS · PROCEDURAL · AI (C)
D	**Material artist** part-time, 2 days / week	Captures samples, reviews every material by eye, builds flagship presets, writes material notes	CAPTURE OPS · ART REVIEW
E	**Founder / product** sales · interviews	Studio interviews, beta programme, pricing, gates, Chaos partner contact, launch	GATES · BETA · LAUNCH

Solo variant. One person covers A + B + C in sequence (46 weeks); hire D from week 8. **Minimum viable team:** A + B — AI can follow v1.0. [Guessing on effort split]

Master schedule — team of three

Lanes run in parallel; dependencies are the only reason anything waits. Thirty calendar weeks, ninety person-weeks.



Critical path: rig + samples → ingest/solve → mapping → capture pipeline → Gate 2. Anything that slips here slips launch. The Inspector lane is independent and should never wait on imaging.

Solo: run lanes A → B → C one after another: 46 weeks, same gates at weeks 10 and ~29. [Guessing on durations — re-plan after Gate 1 with real velocity.]

Inspector MVP, sprint by sprint

Four two-week sprints from an empty repo to a paid beta. Each sprint ends with something a studio can see.

S1 Skeleton

WEEKS 3–4

- pymxs plugin + PySide6 panel via qtmax
- Autodesk .bundle packaging
- Scene walker: VRayMtl, CoronaPhysicalMtl, legacy, bitmaps
- Material data model + JSON export

EXIT Panel lists every material in a real scene

S2 Rules

WEEKS 5–6

- Data maps loaded as sRGB
- Albedo luminance, metal F0, IOR tiers
- Normal-map flip vs height
- Gloss/roughness swap, glass without absorption

EXIT Flags real errors in 10 downloaded scenes

S3 Fixes + report

WEEKS 7–8

- One-click fixes, fully undoable
- Batch mode over a folder of .max files
- Report export per scene
- False-positive test on 50 scenes

EXIT False-positive rate below 5 %

S4 Beta

WEEKS 9–10

- Licence keys + installer
- Crash reporting, logging
- Onboarding: 3-minute video
- 10 studios in beta

EXIT Gate 1: ≥ 5 of 10 would pay

Phases 2–7: inputs, outputs, exit

Every phase has one owner and an exit you can check without arguing.

#	PHASE	OWNER	WEEKS	INPUTS	OUTPUTS	EXIT CRITERION
2	**Parameter mapping**	B	3	Rig, 20 samples, renderer installs	Fitted roughness / metal / glass / SSS tables for V-Ray and Corona	Same map set, same look in both — internal check
3	**Builder + presets**	A · C	6	Mapping tables, reference n,k / IOR / σ	Builder UI; 150 measured presets; Substance / Megascans / Poliigon import	Every preset passes map QA; imports fixed automatically
4	**Capture pipeline**	B · D	8	Rig, ingest, solve	Capture app (iOS / Android RAW), full map set from a real sample	≥ 10 new samples: albedo ΔE ≤ 2, normals and height verified · Gate 2
5	**Glass, SSS, fabric**	B	5	Mapping tables, measured scattering data	Glass families with absorption; skin, marble, wax, leaf; sheen fabrics	Back-lit and slab checks pass in both renderers
6	**AI generator**	C	10	MatSynth CC0 subset, captures (held out)	Text / photo → full map set, seamless, 2–4K	≥ 60 % pass map QA before correction; licence manifest clean
7	**Ship v1.0**	A · E	4	All modules	App Store + own store, licence server, docs, CI matrix	15 artists through blind test; crash-free ≥ 99 %

Capture kit and toolchain

Everything a two-person studio could set up in a day. Prices are rough guides — check current pricing before buying.

CAPTURE KIT

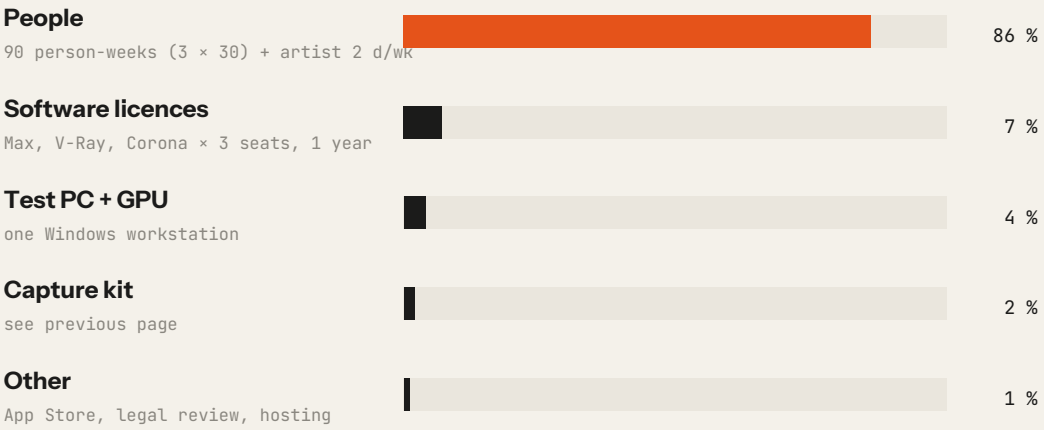
Camera with RAW	mirrorless body or RAW-capable phone	$0–1,200
Lens	fixed 50 mm or macro, manual focus	$200–500
Copy stand or tripod + arm	camera straight down, fixed height	$150–300
LED light + polariser film	one continuous LED, linear film with marked axis	$80–150
Lens polariser	linear, rotatable	$40–80
ColorChecker Classic	24-patch reference, in every frame	$90–120
Chrome + black spheres	light direction and polariser null	$30–60
Scale markers, white card	physical size and flat-field	$20
Colorimeter (optional)	albedo truth at 3+ spots	$500–2,000
Kit total		**≈ $1,100 – 4,400**

TOOLCHAIN

Repository	one monorepo: plugin/ · engine/ · capture-app/ · presets/ · tests/
Plugin	Python 3.11 + pymxs + PySide6 in 3ds Max 2025–2027
Engine	Python + PyTorch in its own process; NumPy / OpenCV for imaging
Capture app	Swift (AVFoundation Bayer RAW) · Kotlin (Camera2 RAW_SENSOR)
Test rig	self-hosted Windows PC with Max + V-Ray + Corona — cloud CI can't run Max
Nightly	50-scene Inspector regression + preset map-QA + both-renderer look check
Licences	3ds Max, V-Ray, Corona per developer; ask Chaos about partner licences
Releases	signed .bundle installer; follow each V-Ray / Corona update within 2 weeks

Budget shape and risk register

People are the budget. Kit and licences are rounding errors next to ninety person-weeks.



Shares are indicative [Guessing]; the absolute figure depends on salaries where you hire. Revenue from the Inspector beta (week 10) should start covering licences before Gate 2.

RISK	TRIGGER → RESPONSE
Chaos builds the same	Chaos announces scene QA or customer scanning → lean on both-renderer support and same-day capture
Renderer update breaks mapping	V-Ray / Corona release → nightly look check fails → re-fit tables within 2 weeks
Gate 1 fails	< 5 of 10 studios would pay → stop capture work, rethink product
Capture too fiddly	Artist takes > 60 min per sample → guided live-view UI, or support a light box
Key person leaves	Imaging engineer leaves → documented solver, paired work from week 4
Training data licence	Unclear source in manifest → drop asset; CC0 + own captures only

From physical sample to finished material

Every material goes down the same line. Steps 3–6 are automatic; people capture and review.



Throughput target: about 40–60 minutes of people time per captured material, so one operator can finish 8–10 materials a day once the line is tuned. [Guessing — measure in Phase 4.]

Other sources join at step 05. Presets, imported Substance / Megascans / Poliigon sets and AI output skip capture but go through the same QA, build, review and publish steps — no shortcut for any source.

Capture session protocol

Ten steps, same order every time. The checklist is printed and taped to the copy stand.

01 **Prepare the sample**
Flat, clean, dry, at least 30 × 30 cm. Note finish and any directionality.

02 **Set the camera**
Straight down. Manual focus locked, base ISO, manual exposure, fixed white balance, RAW only.

03 **Place the references**
ColorChecker, scale marker, chrome sphere and black sphere at the frame edge.

04 **Shoot the flat-field**
White card at the sample plane, once per session, for light fall-off correction.

05 **Light at position 1**
Rotate the light's polariser until the black sphere's highlight disappears. Shoot crossed, then parallel.

06 **Positions 2–8**
Move the light in 45° steps at the same height and distance. Re-null the polariser each time.

07 **Co-located flash frame**
One frame lit from the camera position — this is the frame that measures roughness.

08 **Optional passes**
Rotated light for brushed metal; grazing light for velvet and fabric sheen.

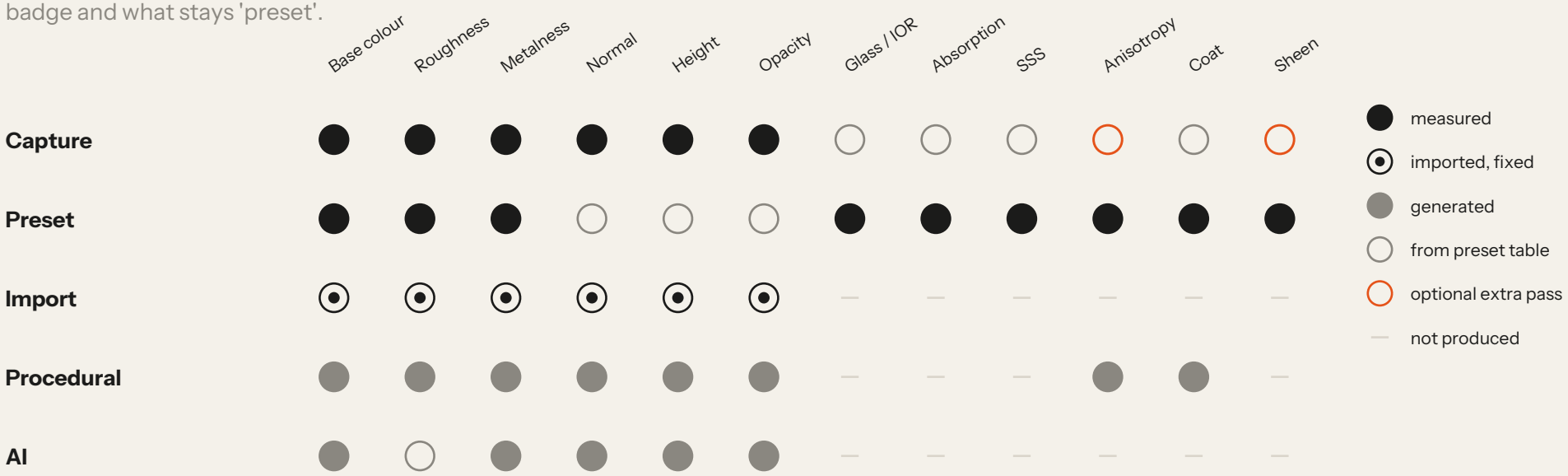
09 **Check the set**
18 frames, nothing clipped, every reference visible. Re-shoot now, not tomorrow.

10 **Log and ingest**
Sample ID, finish, notes. Ingest refuses anything that isn't Bayer RAW.

ELEVATION 45–60° · DISTANCE ≥ 3× SAMPLE · 8 POSITIONS · SAME EXPOSURE ON EVERY FRAME

Which source makes which map

Not every map can be measured with a phone rig. The matrix decides what gets a 'captured' badge and what stays 'preset'.



Format for every map: 16-bit, tileable, 2048 px per metre (4096 for close-ups), physical scale stored. Colour maps sRGB; every data map linear. One structure signal drives colour, roughness and height together.

Badges follow the matrix: a material is **Captured** only for the maps the rig really measured; everything else is labelled **Preset**, **Generated** or **Imported** in its material.json.

Definition of done

A material is finished when it passes twelve checks and a human look — never before.

Draft › **Solved** › **QA passed** › **Built** › **Reviewed** › **Published**

- [ ] 01 Albedo ΔE2000 ≤ 2 against the calibrated photo
- [ ] 02 Albedo luminance 30/50–240 sRGB
- [ ] 03 Metal: ≥ 95 % of texels at 0 or 1
- [ ] 04 Metal colour matches measured F0
- [ ] 05 Roughness has real spread, tied to height detail
- [ ] 06 Roughness and gloss not swapped
- [ ] 07 Normal agrees with height (OpenGL / DirectX checked)
- [ ] 08 Height in millimetres, scale stored
- [ ] 09 Seamless: no wrap seams, no visible repetition
- [ ] 10 Colour space right on every map
- [ ] 11 Glass carries a measured absorption colour
- [ ] 12 Material builds in V-Ray and Corona with native nodes only

Then the human look. The artist views each material on a standard shader ball in both renderers (internal only) and approves or sends it back with a note. Rejections are logged so the rules improve.

How the team works, week to week

Two-week sprints, one weekly material review, and a standing rule: every V-Ray or Corona release is absorbed within two weeks.

MON	TUE	WED	THU	FRI
Stand-up · 15 min	Stand-up · 15 min	Stand-up · 15 min	Stand-up · 15 min	Stand-up · 15 min
Sprint planning WK 1 · 60 MIN	**Capture day** ARTIST + IMAGING	**Build day** NO MEETINGS	**Capture day** ARTIST + IMAGING	**Material review** 45 MIN · EVERY WEEK
				Demo + retro WK 2 · 60 MIN

MEASURED EVERY SPRINT

8+ after Phase 4
Materials published
per week

≥ 80 %
QA first-pass rate
share passing all 12 checks

≤ 2.0
Albedo accuracy
mean ΔE2000 vs instrument

< 5 %
Inspector false positives
on the 50-scene set

≥ 99 %
Crash-free sessions
in 3ds Max

≤ 14
Renderer lag
days to support a new release

TARGETS ARE PROPOSED — SET THE REAL ONES FROM SPRINT 1 VELOCITY AND THE PHASE 4 CAPTURE TIMES.